



Videojet 1000 Line Series WSI Communications Protocol

**P/N 462115 - 01
Revision: AD, July, 2013**

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1. Introduction

This software provides a simple protocol to implement for 1000 Series. It achieves the following specific goals:

- To provide an upgrade path for existing Willett customers with applications utilizing the simpler protocols which existed in earlier Willett products
- To allow communication with PLC's which are not flexible enough to operate more complex protocols
- To provide a "quick and simple" protocol which the Willett customers may use to provide their own integrated solution to a coding or marking application
- To provide a fast, light-weight protocol for use in higher speed printing applications
- To stop the jet remotely

However, this protocol does not support the following highly involved functions which are features of the existing Willett protocol:

- Operation over a modem link in an RS485 multi-drop environment
- Creation or deletion of messages, user fields and logos
- Execution of Jet Start control function

The protocol operates on ports COM1 and COM2 and on a TCPIP stream on ports 3100 (and 3102 for head 2 on a dual head printer), selected via the **REMOTE** option on the **CONFIGURE** menu.

Note: To set the baud rate, number of stop bits, data bits etc. use the **SERIAL PORT BAUD RATES** option on the **CONFIGURE** menu.

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2. Supported Packet Types

The following packet types are supported for inclusion in the simple protocol:

Command Name	Command Number	Function
Message Select	M	Select a message to print from the list of existing messages
Delete Message Text	C	Clear all existing text from the message that is currently printing
Update Message Text	T	Update the message content of the message that is currently printing
Set Message Parameters	P	Set the print parameters associated with the message that is currently printing
Delete User Field Data	D	Clear data from an existing named user field
Update User Field Data	U	Update the data within an existing named user field
Update Logo Data	L	Update the bit pattern of an existing named logo
Set Module Widths	B	Set the module widths (bars and gaps) for printing barcodes
Stop Jet	K	Performs a jet stop sequence
Request Part Number	H	Return the Software Part Number from the machine
Request Error Status	E	Return the errors currently operative within the software
Set Date and Time	Z	Sets the specified Date and Time for printer
Request Current Selected Message	Q	Returns the current selected message name for printing
Print On/Off	O	Turns On or Off printing
Request Print/Product Counter	G	Returns the Print or Product counter value depending on the data sent with the command.
Reset Print/Product Counter	R	Resets the Print or Product counter depending on the data sent with the command

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3. General Protocol Packet Specification

For information on protocol commands for different variants, refer [Chapter 5: Appendix](#)

Each protocol packet sent to the machine must adhere to the following general format:

[STX] [TYPE] [DATA] [ETX]

[STX] Is a fixed single ASCII character 02h which is used to indicate the start of the protocol packet.

[TYPE] Is a single ASCII character in the range 'A' to 'Z' used to describe the type of protocol packet which is being sent.

[DATA] Is a multiple ASCII character variable length field, where all character codes are within the range ASCII 20h to 7Fh plus 0Ah (i.e. no control codes, apart from 0Ah is used as a field separator).

[ETX] Is a fixed single ASCII character 03h which is used to indicate the end of a protocol packet.

In response, the machine will reply with a three character check sequence as follows:

\$XX \$ indicates the command was executed successfully. XX is a pair of ASCII characters within the range ASCII '0' to '9' or 'A' to 'F'. The character pair XX describes the modulo 256 sum of all characters between the [STX] and [ETX] fields, (not including those fields). The first character represents the high four bits of the CheckSum, and the second character the low four bits of the CheckSum.

!XX ! indicates response for unknown command or on failure of a command. XX is a pair of ASCII characters within the range ASCII '0' to '9' or 'A' to 'F'. The character pair XX describes the modulo 256 sum of all characters between the [STX] and [ETX] fields, (not including those fields). The first character represents the high four bits of the checksum, and the second character, the low four bits of the checksum.

3.1. ***“Message Select” Packet Format***

Select a message to print from the list of existing messages stored within the machine. If the message does not exist, the currently printing message will remain unchanged.

[STX] [TYPE] [MESSAGE NAME] [ETX]

- The TYPE field contains the single ASCII character ‘M’.
- The MESSAGE NAME field is a string of between 1 and 30 ASCII characters specifying a message name stored within the machine.

3.2. ***“Delete Message Text” Packet Format***

Clear all existing text from the currently printing message, leaving the message content blank, but the message parameters unchanged. This command does not delete the message.

[STX] [TYPE] [ETX]

- The TYPE field contains the single ASCII character ‘C’.

3.3. ***“Update Message Text” Packet Format***

Updates the message content of the currently printing message in line with the new data supplied within the data field of the packet. This command deletes the current message content before the new text is inserted. The command is processed in synchronous mode – this means that the packet will not be acknowledged until processed completely. In case the new message text includes prompted user fields the packet will not be acknowledged until the operator inputs the prompted text.

[STX] [TYPE] [MESSAGE DATA 1] [SEP] [MESSAGE DATA n] [ETX]

- The TYPE field contains the single ASCII character ‘T’.
- The MESSAGE DATA fields contain several sub fields as described below. There may be several MESSAGE DATA fields within a single message, each separated by a SEP field.
- The SEP field may only contain the single ASCII characters LF (0Ah), and is used to separate data fields within a message data packet.

3.3.1. MESSAGE DATA Field Format

Each message data field is of the following format:

[FONT NUM] [HORC] [VERC] [ATTRIB] [MESSAGE TEXT]

- The FONT NUM field consists of exactly 2 characters (ASCII '0' to '9') which represent a decimal value between 1 and 99. It describes the font number to be used for the following text. The font number follows the order in which the fonts appear when using the editor, i.e. font 0 is always 7hi, but the rest will be dynamically allocated according to the fonts fitted. (**00 = 7hi, 01 = 9hi, 02 = 12hi, 03 = 16hi, 04 = 24hi, 05 = 34hi, 06 = 5hi**).
- The HORC field consists of exactly 4 characters (ASCII '0' to '9') which represent a decimal value between 0 and 9999, it describes the relative horizontal position at which the text fragment should be placed within the message. **Note: This position is relative, and subject to left justification.** It should be considered as a fragment ordering control rather than a means of placing the text fragment in an exact location; i.e., the text fragment with a HORC of 1 will always be to the left of another text fragment with a HORC of greater than 1 and more. The numeric data within the packet field is right justified and zero-padded.
- The VERC field consists of exactly 3 characters (ASCII '0' to '9') which represent a decimal value between 0 and 999. It describes the vertical position at which the text field should be placed within the message. **Note: The whole message is subject to upward justification.** If blank lines are required at the top of a message, a dummy fragment containing a single space character in an appropriate font must be sent. The numeric data within the packet field is right justified and zero-padded. The algorithm of field positioning on the basis of HORC and VERC is as follows:
 1. Select all fields with minimal value of HORC.
 2. Place them to the message image in a left aligned column from top to bottom in the order of VERC values with 1 pixel gap.
 3. Select all fields with second minimal value of HORC and place them in same column to the right of the fields with minimal HORC.
 4. Shift each fields of the second column horizontally to the left until there is a minimal gap of 1 pixel with any of the fields of the first column or it comes to the left boundary of the message image.
 5. Continue selecting fields with next value of HORC, piling them in a column and shifting left where possible, until all the fields take their place.

- The ATTRIB field consists of exactly 6 characters (ASCII '0' to '9' or 'A' to 'F'). The first four characters represent a hexadecimal value between 0 and FFFF, it describes a bit mask which is used to apply various attributes to the message text field. The sixth character represents a hexadecimal value between 0 and F, it describes the code page number to be used. To clarify, this field should be into 3 sub-fields, in the order of their appearance:
 - 3 attribute characters. Each attribute is a bit mask, in which the 4 least significant bits are considered to yield an integer valued between 0-15, the base 16 (hexadecimal) interpretation of this integer (0-F), is then encoded into an ASCII character from the range '0'-'9', 'A'-'F'.

The bits in each attribute are set according to whether the corresponding properties are set, as follows:

ATTRIBUTE 1

Bit No	Attribute
Bit 1	INVERT
Bit 2	REVERSE
Bit 3	CLOCK_CODE
Bit 4	Double Dots On

ATTRIBUTE 2:

Bit No	Attribute
Bit 1	Triple Dots On
Bit 2	TOWER_PRINTING
Bit 3	PROMPTED_USER_FIELD
Bit 4	BARCODE_HR_ON

ATTRIBUTE 3:

Bit No	Attribute
Bit 1	Use Custom Font
Bit 2	Inverse Video
Bit 3	BARCODE_CHECKSUM_ON
Bit 4	Not Used

- Barcode type: This is a value between 0–25 (see table below for definition). This is encoded as 2 ASCII characters in the range '0'-'9', 'A'-'F' as follows: The first character is formed by dividing the required barcode type by 16 and converting the base 16 interpretation of the dividend into an ASCII character (of range '0'-'9', 'A'-'F'). The second character is created in a similar fashion, except that it is the remainder of the division by 16 process whose base 16 interpretation is then converted to an ASCII character (of range '0'-'9', 'A'-'F').

- The barcode is elected from the following range:

ATTRIBUTE 4 & 5:

Value	Attribute
00	Barcode None
01	ITF
02	EAN 8
03	EAN 13
04	UPC-A
05	UPC-E
06	Code-128
07	EAN 128
08	Code 39
09	Data Matrix 10 x 10
10	Data Matrix 12 x 12
11	Data Matrix 14 x 14
12	Data Matrix 16 x 16
13	Data Matrix 18 x 18
14	Data Matrix 20 x 20
15	Data Matrix 22 x 22
16	Data Matrix 24 x 24
17	Data Matrix 26 x 26
18	Data Matrix 32 x 32
19	Data Matrix 8 x 18
20	Data Matrix 8 x 32
21	Data Matrix 12 x 26
22	Data Matrix 12 x 36
23	Data Matrix 16 x 36
24	Data Matrix 16 x 48
25	GS1 Databar

- The code page number, whose value in base 16 ranges from 0 to F. This is encoded as a single ASCII character (of range '0'-'9', 'A'-'F').
Note: 1510/1520 and 1610/1620 printers will ignore this code page number input.
- The MESSAGE TEXT field consists of up to 200 ASCII characters which describe the text to be entered within the message, or (if it starts with the character "@") the name of a single user field. This arrangement allows plain text, barcodes, counters, or other real-time data to be inserted within the message. The print orientation and presentation is controlled via the attribute field.

3.3.2. An Example of Update Message Text

Print a twin line message:	Top line (7high font) with data "12345"
	Bottom line (9high font) with data "ABCDE"
	Reverse characters on

Data: (spaces provided to separate fields, do not actually include spaces in data)

**[02]T 00 0001 034 200000 12345[0A] 01 0001 025 200000
ABCDE [03]**

T = Command name	[0A] = separator
00 = 7 high font	01 = 9 high font
0001 = HORC	0001 = HORC
034 = VERC	025 = VERC (34 – 7high font – 2 drops of space)
200000 = ATTRIB (reverse print on)	200000 = ATTRIB (reverse print on)
12345 = DATA	ABCDE = DATA

3.4. “Modify Message Parameters” Packet Format

Update the print parameters associated with the currently printing message, in line with the new parameter list supplied within the data field of the packet.

**[STX] [TYPE] [REV] [INV] [WID] [EHT] [GAP] [EXP] [HEJRA][DLY] [BLD] [DRP]
[RASSUB][RLEN][RAS] [RDLYLEN] [RDLY] [ETX]**

- The TYPE field contains the single ASCII character ‘P’.
- The REV field consists of a single ASCII character where ‘0’ represents OFF and ‘1’ represents ON. It describes the message parameter “message reverse”.
- The INV field consists of a single ASCII character where ‘0’ represents OFF and ‘1’ represents ON. It describes the message parameter “message invert”.
- The WID field consists of exactly 4 characters (ASCII ‘0’ to ‘9’) which represent a decimal value ranged as indicated in table. It describes the message parameter “width”. The value is right-justified and zero-padded within the field. Unit of this field is depending on the “Display Unit of

measure” & “Message Configuration” parameters in printer. In Case measuring unit is selected as Inches, unit of this field will be 1/100th of inch. For example, 1.25” should be sent as 125. In case of Metric units, it will be in mm. Please refer following table for value range:

Metric (mm)		1/100 Inches	
Divider / Strokes	Linear Units	Divider / Strokes	Linear Units
1-255	1-999	1-255	1-3936 (1=0.01”)

When Shaft Encoder source is selected as Auto Encoder

Metric (mm)		1/100 Inches	
Strokes per mm	Linear Units	Strokes per Inches	Linear Units
1-10	1-999	1-254	1-3936 (1=0.01”)

- The EHT field consists of exactly 2 characters (ASCII ‘0’ to ‘9’) which represent a decimal value between 1 and 10. It describes the message parameter “message height”. The value is right-justified and zero-padded within the field.
- The GAP field consists of a single character (ASCII ‘0’ to ‘9’) which represents a decimal value between 0 and 9. It describes the message parameter “inter-character gap”.
- The EXP field consists of exactly 5 characters (ASCII ‘0’ to ‘9’) which represent a decimal value between 0 and 32767, it describes the message parameter “expiry date - days ahead counter” which is used in 1210/1220/1510/1520 only. The value is right-justified and zero-padded within the field. In 1610/1620/1710/1610DH this parameter is ignored (left for compatibility) and replaced with Expiry offset in Update User Field packet for date user field.
- The HEJRA field consists of exactly 5 characters (ASCII ‘0’ to ‘9’) which represent a decimal value between 0 and 32767. It describes the message parameter “Hejra expiry days ahead” which is used in 1210/1220/1510/1520 only. The value is right-justified and zero-padded within the field.
- The DLY field consists of exactly 5 characters (ASCII ‘0’ to ‘9’) which represent a decimal value ranged as indicated in table. It describes the message parameter “product delay”. The value is right-justified and zero-padded within the field. Unit of this field depends on the “Display Unit of measure” & “Message Configuration” parameters in printer. In Case measuring unit is selected as Inches, unit of this field will be 1/100th of inch. For example, 1.25” should be sent as 125. In case of Metric unit, it will be in mm.

Metric (mm)		1/100 Inches	
Divider / Strokes	Linear Units	Divider / Strokes	Linear Units
0-9999	0-999	0-9999	0-3936 (1 = 0.01")

- The BLD field consists of exactly 2 characters (ASCII '0' to '9') which represent a decimal value between 1 and 10. It describes the message parameter "raster repeat". The value is right-justified and zero-padded within the field.
- The DRP field consists of exactly 2 characters (ASCII '0' to '9') which represent a decimal value between 1 and 34. It describes the message parameter "printed dots". The value is right-justified and zero-padded within the field. The field is available for 1510/1520 only.
- The RASSUB field consists of a single ASCII character where '0' represents OFF and '1' represents ON. It describes the message parameter "Raster Substitution".
- The RLEN field consists of exactly 3 characters (ASCII '0' to '9') which represent a decimal value between 1 and 255. It describes the length of the raster filename that is to follow. The value is right-justified and zero-padded within the field.
- The RAS field consists of ASCII characters as per the previous RLEN field. It describes the message parameter "Raster".
- The RDLYLEN field consists exactly 1 (ASCII '0' to '9') character which represents the Length of the Reverse Print Delay Value. It sets the Reverse Product Delay length to support a variable length Reverse Product Delay in 1510/1520 and 1610/1620 only. The field is not required if the reverse delay length is fixed.
- The RDLY field consists of exactly 5 characters (ASCII '0' to '9') which represent a decimal value ranged as indicated in table. It describes the message parameter "reverse product delay". Unit of this field is depending on the "Display Unit of measure" and "Message Configuration" parameters in printer. In Case measuring unit is selected as Inches, unit of this field will be 1/100th of inch. For example 1.25" should be sent as 125. In case of Metric unit, it will be in mm.

Metric (mm)		1/100 Inches	
Divider / Strokes	Linear Units	Divider / Strokes	Linear Units
0-9999	0-999	0-9999	0-3936 (1 = 0.01")

3.4.1. An Example of “Modify Message Parameters Command”

Data: (spaces provided to separate fields, do not actually include spaces in data)
This example is for 15xx/16xx.

Note: *Expiration Days and Hejra Days is disabled in 16xx but still needs to be sent along with CMD.*

[02] P 0 0 0350 06 2 00031 00001 00100 01 16 1 007 16-high 5 00350 [03]

P = command	00100 = Delay (1 inch or 100mm)
0 = Reverse message off	01 = Raster Repeat
0 = Invert message off	16 = Printed drops
0350 = Width (3.5 inches or 350mm)	1 = Raster substitution ON
06 = Character Height	007 = RLEN is 7 Characters
2 = Character gap	16-high = Raster name (enter name as it appears on the screen)
00031 = Expiration days ahead	5 = RDLYLEN is 5 character of reverse product delay
00001 = Hejra Expiration	00350= RDLY Reverse Product Delay

3.4.2. An Example of “Modify Message Parameters Command”

Data: (spaces provided to separate fields, do not actually include spaces in data)

[02]P 0 0 0350 07 1 00060 00000 00100 01 16 1 007 16-high [03]

P = command	00000 = Hejra Expiration
0 = Reverse message off	00100 = Delay (1 inch or 100mm)
0 = Invert message off	01 = Raster Repeat
0350 = Width (3.5 inches or 350mm)	16 = Printed drops
07 = Character Height	1 = Raster substitution ON
1 = Character gap	007 = RLEN is 7 Characters
00060 = Expiration days ahead	16-high = Raster name (enter name as it appears on the screen)

3.5. “Clear User Field Data” Packet Format

Clear data from an existing named user field, leaving the user field blank. This command does not delete the user field.

[STX] [TYPE] [USER FIELD NAME] [ETX]

- The TYPE field contains the single ASCII character ‘D’.
- The USER FIELD NAME field is a string of between 1 and 30 ASCII characters describing an existing user field stored within the machine.

3.6. “Update User Field Data” Packet Format

Update the data within an existing named user field in line with the new data supplied within the data field of the packet. This command deletes the current user field content before the new text is inserted. User fields of type: text, serializer and date/time can be configured using this message.

[STX] [TYPE] [USER FIELD NAME] [SEP] [USER FIELD DATA] [ETX]

- The TYPE field contains the single ASCII character ‘U’.
- The USER FIELD NAME field is a string of between 1 and 30 ASCII characters describing an existing user field stored within the machine.
- The SEP field may only contain the single ASCII characters LF (0Ah), and is used to separate data fields within a user field data packet.
- The USER FIELD DATA field is a string of between 1 and 50 ASCII characters describing the new value to be placed within the named user field. The table below describes the presentation used for the fields.
 - SERIALIZER – USER FIELD DATA

Field name	Values	Minimum/Maximum
Start	ASCII [‘0’-‘9’, ‘A’-‘Z’]	Field may vary between 1 and 9 characters width
Separator	0Ah	
End	ASCII [‘0’-‘9’, ‘A’-‘Z’]	Field may vary between 1 and 9 characters width
Separator	0Ah	
Step	ASCII [‘0’-‘9’]	Value between 1 and 100
Separator	0Ah	
Direction	ASCII [‘0’- decrement, ‘1’ – increment]	
Separator	0Ah	

Field name	Values	Minimum/Maximum
Repeat	ASCII ['0'-'9']	Value between 1 and 250
Separator	0Ah	
Padding	ASCII ['0'] or SPACE or EMPTY	Field may vary between 0 and 1 characters width
Separator	0Ah	

Since a separator delimits each field, no justification or padding rules are applied to the serializer format.

- Text
Text field information is sent verbatim, without markup or padding.
- Date/time
Date/time information is formatted in the following way:

Field name	Values
OFFSET_TYPE Note: The field is not valid for 1210/1220/1510/1520 printers. The offset is determined by message parameters.	Single ASCII character as follows: 'N' - DT_OFFSET_NONE 'D' - DT_OFFSET_DAY 'W' - DT_OFFSET_WEEK 'M' - DT_OFFSET_MONTH 'Y' - DT_OFFSET_YEAR
OFFSET_VAL Note: The field is not valid for 1510/1520 printers.	A decimal value between 0-999, represented as ASCII characters (0-9), right justified with zero padding.
CLOCK_FORMAT	ASCII text field, with a maximum of 46 characters.

3.7. “Update Logo Data” Packet Format

Update the bit pattern of an existing named logo in line with the new data supplied within the data field of the packet. This command deletes the current logo content before the new bit pattern is inserted. The image sent must have the same dimensions as the image on the printer.

[STX] [TYPE] [LOGO NAME] [SEP] [NUM DROPS] [NUM RASTERS] [LOGO DATA] [ETX]

- The TYPE field contains the single ASCII character 'L'.
- The LOGO NAME field is a string of ASCII characters describing an existing user field stored within the machine. The maximum size of the field is 30 characters.
- The SEP field contains a single linefeed (0Ah) character used to terminate the logo name field.
- The NUM DROPS field specifies the number of drops per vertical stroke in the logo data which follows. It is a decimal value between 5-34 encoded as 2 ASCII characters in the range '0'-'9'. The data is right-justified and zero-padded.
- The NUM RASTERS field specifies the width of the logo as a number of rasters. It is a decimal value between 1 and 255 encoded as 3 ASCII characters in the range '0'-'9', the data is right-justified and zero-padded.
- LOGO DATA is encoded as a list of pairs of ASCII characters, whereupon each pair represents 8-bits of image data. The process in which the graphic is converted into ASCII characters is not trivial and is as follows:
 - The image is broken down into rasters NUM DROPS high. There are NUM RASTERS quantity of rasters.
 - A raster is depicted as a row of ink-drop data, in which each drop position may or may not be "inked", and each drop is assigned a zero-based drop number. The drop number ascends from the top of the image – drop number 0 being at the top.
 - To encode each raster, a bit array is created whose size is either equal to NUM DROPS if NUM DROPS is multiple of 8, or the next highest multiple of 8 if not.
 - The state of each drop, whether it is inked or not, in the raster results in the corresponding bit in the array being set or cleared, respectively.
 - Each bit is set/cleared from the lowest bit, corresponding to the topmost drop. Where the array size is not equal to NUM DROPS, the bit position in the array populated is equal to the drop number offset by the addition of the constant array size – NUM DROPS. Any unused bits should be left clear.
 - As each raster is coded into a bit array, starting at the least significant bit, each group of 4-bits is interpreted as a base-16 number (0-F), and this number is converted into the corresponding ASCII character (in range '0'-'9', 'A'-'F').
 - Each ASCII character as it is produced is appended onto the LOGO DATA buffer.

- The process is repeated for each raster from left to right, until there are no more rasters.

Example logo and encoding

		Byte Offset												
		00	02	04	06	08	10	12	14	16	18	20	22	24
Byte n	Bit 7													
	Bit 6			0	0	0	0	0	0	0	0			
	Bit 5		0	0	0	0	0	0	0	0	0	0		
	Bit 4	0	0	0							0	0	0	
	Bit 3	0	0								0	0	0	
	Bit 2	0	0							0	0	0	0	
	Bit 1	0	0						0	0	0	0	0	
	Bit 0	0	0					0	0	0		0	0	
Byte n+1	Bit 7	0	0				0	0	0			0	0	
	Bit 6	0	0			0	0	0				0	0	
	Bit 5	0	0		0	0	0					0	0	
	Bit 4	0	0	0	0	0						0	0	
	Bit 3	0	0	0	0							0	0	
	Bit 2	0	0	0							0	0	0	
	Bit 1		0	0	0	0	0	0	0	0	0	0		
	Bit 0			0	0	0	0	0	0	0	0			

The above logo would be represented by the following data:

0x1F, 0xFC, 0x3F, 0xFE, 0x70, 01F, 0x60, 0x3B, 0x60, 0x73, 0x60, 0xE3, 0x61, 0xC3, 0x63, 0x83, 0x67, 0x03, 0x7E, 0x07, 0x3F, 0xFE, 0x1F, 0xFC, 0x00, 0x00

It would be transmitted to the machine using the following packet:

<STX>L16 High Zero Logo<LF>160131FFC3FFE701F603B607360E
361C3638367037E073FFE1FFC0000<ETX>

3.8. “Set Module Widths” Packet Format

Set the module widths (bars and gaps) for printing barcodes, to ensure that barcodes are readable regardless of the ink spread on the substrate.

[STX] [TYPE] [BW1] [BW2] [BW3] [BW4] [GW1] [GW2] [GW3] [GW4][ETX]

- The TYPE field contains the single ASCII character ‘B’.
- The BW1 field consists of exactly 2 characters (ASCII ‘0’ to ‘9’) which represent a decimal value between 1 and 12. It describes the number of rasters in the narrowest bar of the printed raster.
- The BW2 field consists of exactly 2 characters (ASCII ‘0’ to ‘9’) which represent a decimal value between 1 and 12. It describes the number of rasters in the next narrowest bar of the printed raster.
- The BW3 field consists of exactly 2 characters (ASCII ‘0’ to ‘9’) which represent a decimal value between 1 and 12. It describes the number of rasters in the next to widest bar of the printed raster.
- The BW4 field consists of exactly 2 characters (ASCII ‘0’ to ‘9’) which represent a decimal value between 1 and 12. It describes the number of rasters in the widest bar of the printed raster.
- The BG1 field consists of exactly 2 characters (ASCII ‘0’ to ‘9’) which represent a decimal value between 1 and 12. It describes the number of rasters in the narrowest gap of the printed raster.
- The BG2 field consists of exactly 2 characters (ASCII ‘0’ to ‘9’) which represent a decimal value between 1 and 12. It describes the number of rasters in the next narrowest gap of the printed raster.
- The BG3 field consists of exactly 2 characters (ASCII ‘0’ to ‘9’) which represent a decimal value between 1 and 12. It describes the number of rasters in the next to widest gap of the printed raster.
- The BG4 field consists of exactly 2 characters (ASCII ‘0’ to ‘9’) which represent a decimal value between 1 and 12. It describes the number of rasters in the widest gap of the printed raster.

3.9. “Stop Jet” Packet Format

Invoke a clean jet stop sequence on the printer. If unable to perform clean stop, quick stop will be performed.

[STX] [TYPE] [ETX]

- The TYPE field contains the single ASCII character ‘K’.

3.10. “Request Part Number” Packet Format

Return the “Software Part Number” string stored within the machine.

[STX] [TYPE] [ETX]

- The TYPE field contains the single ASCII character ‘H’.

The machine responds with the following packet:

[STX] [PART NUMBER] [ETX]

- **Example:** Response will be 31 2E 30 2E 33 38 35 57 20 20 20 20 20 20 20 20 which is "1.0.385W(8 Spaces)".

3.11. “Error Status Request” Packet Format

Return the error and alarm conditions currently asserted in the printer.

[STX] [TYPE] [ETX]

- The TYPE field contains the single ASCII character ‘E’.

The machine responds with the following packet:

[STX] [EEEEEE][A] [ETX]

- The EEEEE field contains the 6 byte hexadecimal error status as follows:

Bit	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6
0	Charge error	Pump fault	BAD NOZZLE	Phasing threshold at maximum - no good phasing.\n Flush nozzle and check phase and charge voltages.	Rasters memory overflow detected!\nOne or more rasters must be removed to fix the problem.	Date/Time not set

1	EHT Trip	Cabinet too hot	Mod driver chip over temperature.\n Check mod circuit and/or replace nozzle.	Auto mod failed to obtain good phasing.\n Flush nozzle and check phase and charge voltages.	Valve error - contact service dept.	New Ink Core has a different ink reference.
2	Gutter fault	Ink core service overdue	Fatal error: no phase response from firmware. Contact customer services.	Initial phasing trim failed.\n Flush nozzle and check phase and charge voltages.	Core not filling. Check for leaks, stuck valves etc.	EHT calibration required
3	Ink core empty	Unable to control viscosity	Phasing threshold at minimum - no good phasing.\n Flush nozzle and check phase and charge voltages.	Modulation readback failed.\n Check mod circuit and/or replace nozzle.	Insufficient ink to fill core.\n Install another cartridge and repeat.	N/A

- The A field contains the 7 bit hexadecimal alarm status word as follows:

Bit Number	Alarm Status Condition Represented
3	Alarm Activated – NOT USED
2	Red Trafficator On
1	Amber Trafficator On
0	Green Trafficator On

3.12. “Set Date and Time” Packet Format

Sets the specified Date and Time for the printer.

[STX] [TYPE] [YY] [MM] [DD] [hh] [mm] [ss] [ETX]

- The TYPE field contains the single ASCII character 'Z'.
- The YY field consists of exactly 2 characters (ASCII '0' to '9') which represent a decimal value between 06 and 99(2000 and 2099). It describes the last two digits of the year.
- The MM field consists of exactly 2 characters each (ASCII '0' to '9') which represent a decimal value between 1 and 12. It describes month in digits.

- The DD field consists of exactly 2 characters each (ASCII '0' to '9') which represent a decimal value between 1 and 31. It describes days in digits.
- The hh field consists of exactly 2 characters each (ASCII '0' to '9') which represent a decimal value between 0 and 23. It describes hours in digits.
- The mm field consists of exactly 2 characters each (ASCII '0' to '9') which represent a decimal value between 0 and 59. It describes minutes in digits.
- The ss field consists of exactly 2 characters each (ASCII '0' to '9') which represent a decimal value between 0 and 59. It describes seconds in digits.

3.13. ***“Request Current Selected Message” Packet Format***

Return the “Current selected message name” for printing

[STX] [TYPE] [ETX]

- The TYPE field contains the single ASCII character 'Q'.

The machine responds with the following packet:

[STX] [MESSAGE NAME] [ETX]

- The MESSAGE NAME field contains the name of the message selected in ASCII format.

3.14. ***“Print On/Off Packet Format”***

- Turns On or Off printing. That is

O0 = print off

O1 = print on

Note: Printhead Jet needs to be On for this command to operate correctly.

[STX] [TYPE] [ETX]

- The type field contains the single ASCII character 'O'.

3.15. ***“Print/Product Counter Request” Packet Format.***

Returns the Print or Product counter value depending on the Data sent with the command. To request for the Print counter value Data ‘A’/‘a’ is used, and to get the product counter data ‘B’/‘b’ is used.

Each command to request the counters is of the following format.

[STX] [TYPE] [DATA] [ETX]

- Where the TYPE is command ‘G’- for requesting of counters.
- And DATA can be ‘A’/‘a’ or ‘B’/‘b’ depending on the Print and Product Counter.

The machine responds with the following Packet:

[STX] [X X X X X X X X X X] [ETX]

- Where ‘X’ represents ASCII characters within the range ASCII ‘0’ to ‘9’.The X is the Value of the Requested Counter represented in ten individual digits.

Example1: Get Print Count

Send: [STX]GA[ETX]

Receive: [STX]00000678[ETX]

Example2: Get Product Count

Send: [STX]GB[ETX]

Receive: [STX]00000679[ETX]

3.16. “Print/Product Counter Reset” packet format

This command resets the Print or Product counter depending on the data sent with the command. To reset the Print counter value Data ‘A’/‘a’ is used, and to reset the Product counter data ‘B’/‘b’ is used.

Each command to reset the counter is of the following format:

[STX] [TYPE] [DATA] [ETX]

- Where the TYPE is command ‘R’- for resetting of the counters.
- And DATA can be ‘A’/‘a’ or ‘B’/‘b’, depending on the counter requested.

Example1: Set Print Count

Send: [STX]RA[ETX]

Receive: \$93

Example2: Set Product Count

Send: [STX]RB[ETX]

Receive: \$94

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4. Differences between 1000 series WSI protocol and Videojet/Willett 40/400 series WSI protocol

4.1. *Update Message Text*

- The VERC field is justified to the top of the screen. The value for the upper most line is 034. If you have a message using a 7 high single line, you need to put 034 in the VERC field to get it in the top position. If you need to add a line beneath that is 7 high line, the next VERC value would need to be 026 or lower (34 - 7 high font – 1 space between lines).
- The ATTRIB field is now 6 characters as opposed to 5 on the 40 series. Codes being sent from an external source to create the message will need the extra nibble of information to be accepted as new data. This is because of the addition of 2D codes to the ATTRIB field.

4.2. *Message Parameters*

- Since the 1000 series has different message parameters than the 40 series, this command is different. Fields like RASSUB, RLEN, RDLY, clock attribute offsets and RAS are not included in the 40 series printers, but are necessary for the 1000 series printers.

4.3. *Error Status Request*

- The list of possible errors on the 1000 series is different than the 40 series printers. This results in different bit masks that correspond to the different types of faults and warnings. If an external controller is monitoring the faults and warnings, the code will need to be changed to properly identify the faults. The traffic light status character is the same as the 40 series, so if you are monitoring only the traffic light status, the code need not be changed.

4.4. *Stop Jet*

- The 1000 series has a “Stop Jet” command that will perform a jet stop when sent. This is not available on the 40 series printers.

4.5. *Request Part Number*

- The response from the machine will be different because the printer part numbers are different. It is important to note that the number of characters sent back should be 16. If the printer's part number is less than 16, the response will be padded with spaces.

- The command response !XX was not available in simple WSI protocol. Only \$XX response was available. '\$' indicates that the command was executed successfully and '!' indicates the response for unknown command or on failure of a command.

4.6. Network Settings

Setting	1510/20	1610/20	1710	1610DH
Maximum Message Length	6000 strokes or 500 characters (Whichever comes first)	10000 strokes or 500 characters (Whichever comes first)	10000 strokes or 500 characters (Whichever comes first)	10000 strokes or 500 characters (Whichever comes first)
Ethernet Port No.	3100	3100	3100	3100
IP Address	192.168.100.002	192.168.100.002	192.168.100.002	192.168.100.002
Subnet Setting	255.255.255.000	255.255.255.000	255.255.255.000	255.255.255.000
Default Gateway setting	000.000.000.000	000.000.000.000	000.000.000.000	000.000.000.000

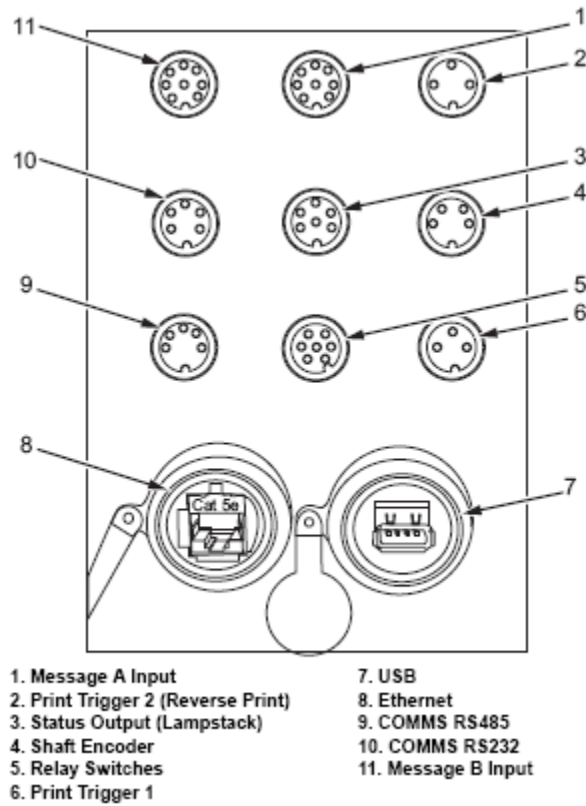
Do the following tasks to change the Network settings:

1. Log into the user interface (UI) at level 3 password.
2. Navigate to **Configure > Network settings**
3. Select the required setting to change.
4. Delete the existing information and enter the new data.

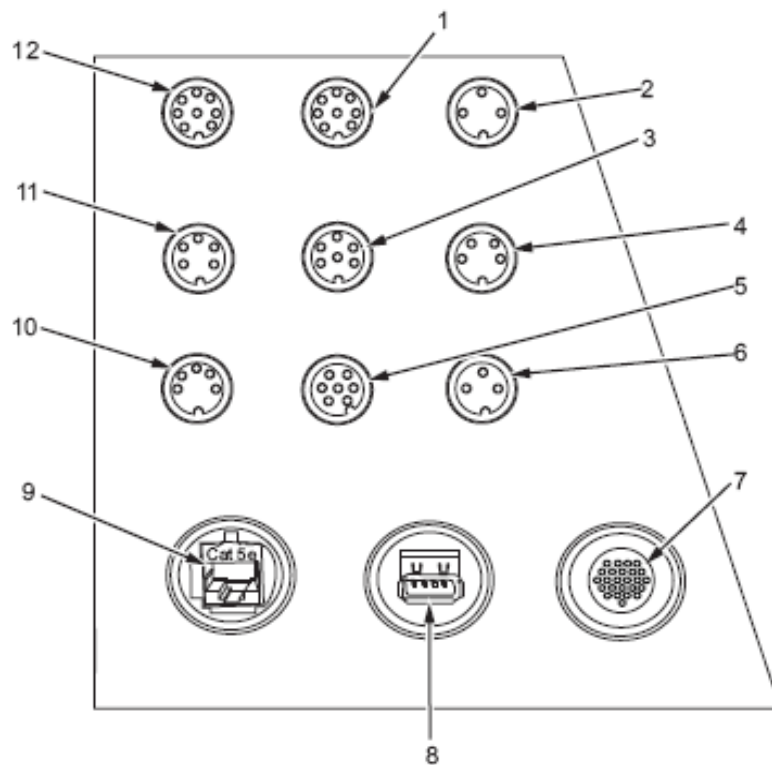
Note: When DHCP is selected, the network assigns the IP address automatically.

4.7. Connectors and Cables

4.7.1. Videojet 1210/1220/1510/1520

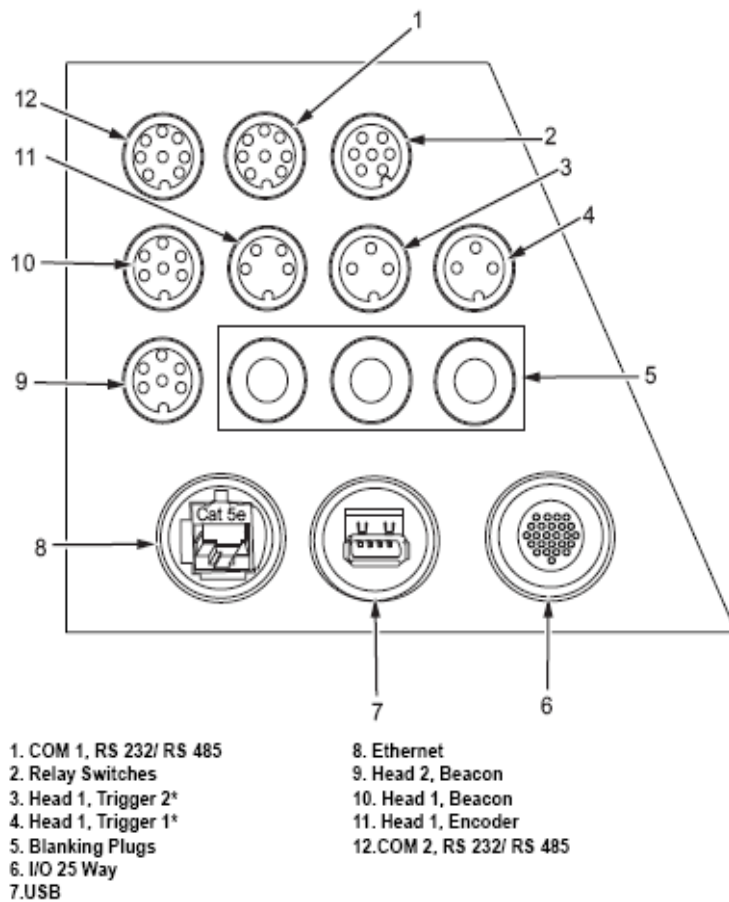


4.7.2. Videojet 1610/1620



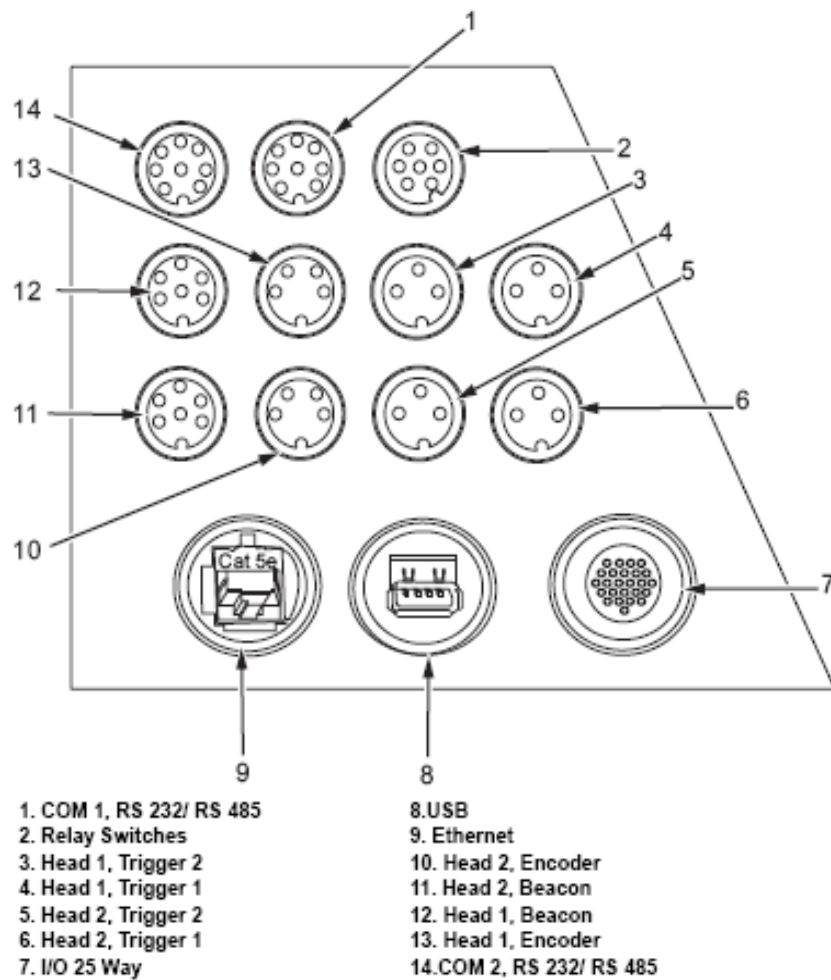
- | | |
|------------------------------------|---------------------|
| 1. Message A Input | 7. I/O 25 Way |
| 2. Print Trigger 2 (Reverse Print) | 8. USB |
| 3. Status Output (Lampstack) | 9. Ethernet |
| 4. Shaft Encoder | 10. COMMS RS485 |
| 5. Relay Switches | 11. COMMS RS232 |
| 6. Print Trigger 1 | 12. Message B Input |

4.7.3. Videojet 1610 DH (S-Mode)

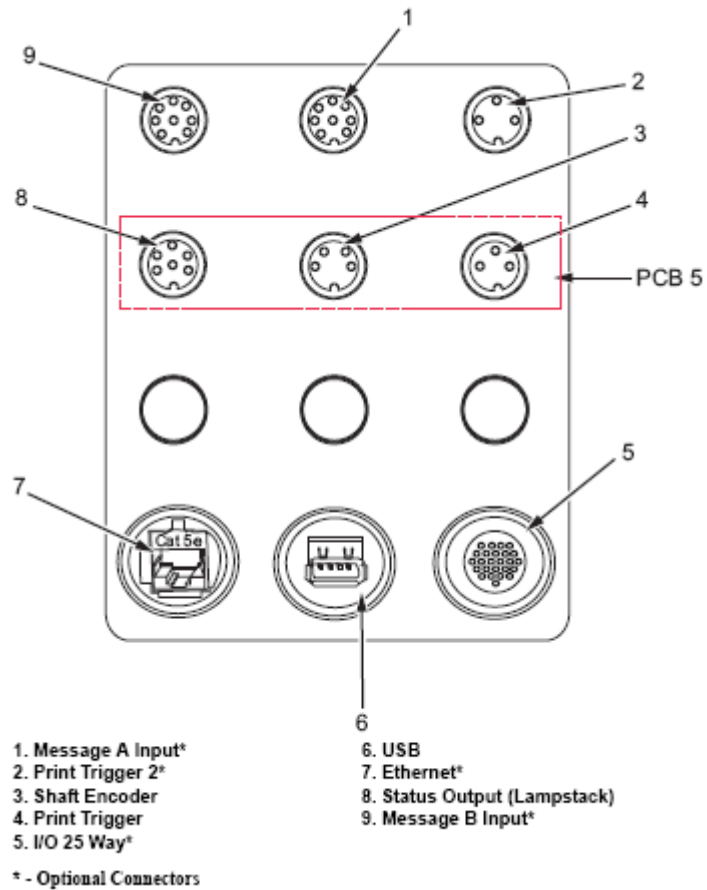


* - The connectors are used for both Head 1 and Head 2 Trigger, though it is labeled as Head 1, Trigger 1 and Head 1, Trigger 2.

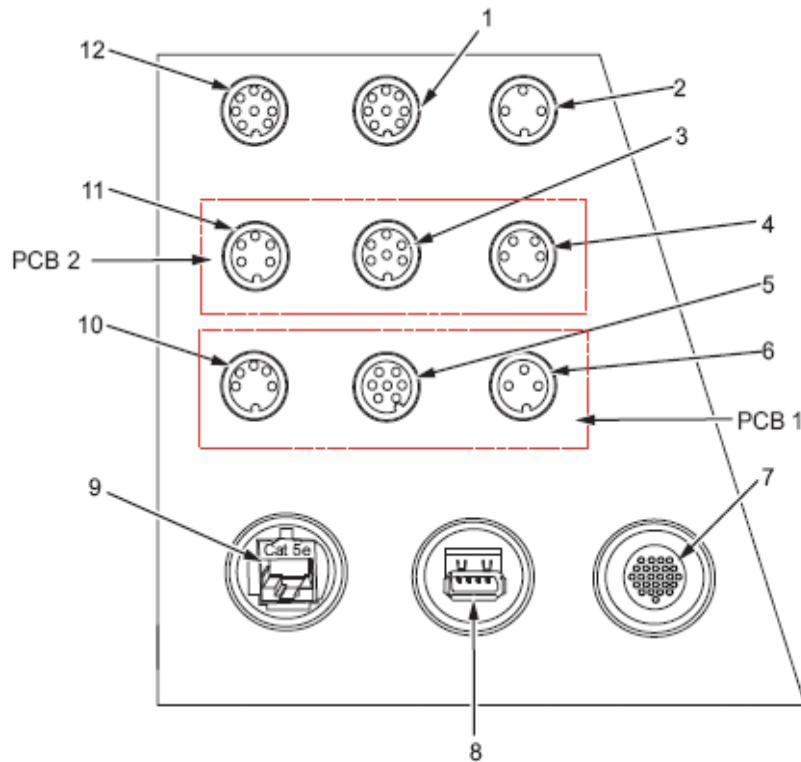
4.7.4. Videojet 1610 DH (I-Mode)



4.7.5. Videojet 1710 (IP 55)



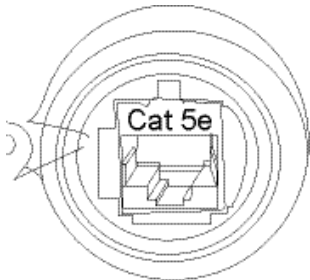
4.7.6. Videojet 1710 (IP 65)



- 1. Message A Input*
 - 2. Print Trigger 2 (Reverse Print)*
 - 3. Status Output (Lampstack)
 - 4. Shaft Encoder
 - 5. Relay Switches
 - 6. Print Trigger 1
 - 7. I/O 25 Way*
 - 8. USB
 - 9. Ethernet*
 - 10. COMMS RS485
 - 11. COMMS RS232
 - 12. Message B Input*
- * - Optional Connectors

Ethernet Connector

The Ethernet connector is used to connect the printer to a remote computer or network to import data or control the printer remotely.



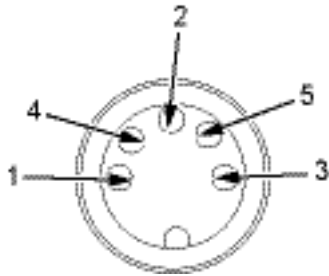
Part Number

External Ethernet connector cable 5 meter: VJ P/N: SP223213

Ethernet connector kit spares VJ P/N: SP399083

RS-485 Connector

The RS-485 connector is a 5 way DIN connector. Use this to connect the printer to an RS-485 network to import data or control the printer remotely.



Note: This view is of the connector at the side of the printer front view.

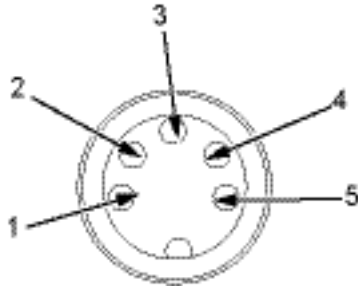
Pin	Function
DIN Pin 1	Line A
DIN Pin 2	Line B
DIN Pin 3	0 VDC Common
DIN Pin 4	-
DIN Pin 5	-

Part Number

Connector for (RS485) male DIN 5 pin connector 180 degree stereo VJ P/N: 234433

RS-232 Connector

The printer has an RS-232 communication port linked to the COMM1 connector. The RS-232 connector is a 5 way DIN connector. The pin allocation is shown in table.



Note: This view is of the connector at the side of the printer front view.

Pin	Function
DIN Pin 1	0 VDC Common
DIN Pin 2	Transmit data out from the printer.
DIN Pin 3	Receive data into the printer.
DIN Pin 4	RTS output from the printer.
DIN Pin 5	CTS input to the printer.

Part Number

Connector, for (RS232) male DIN 5 pin connector VJ P/N: 500-0036-582

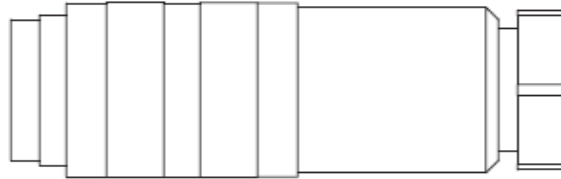
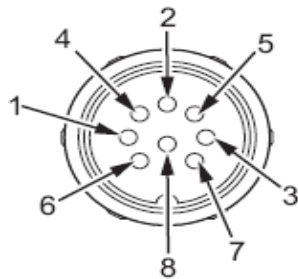
Extension cable, 6 meter with DB9 on host / PC side: VJ P/N: 399062*

Serial cable, 3 meter with DB9 on host / PC side: VJ P/N: 80200258

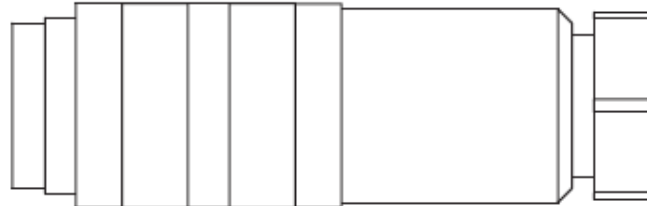
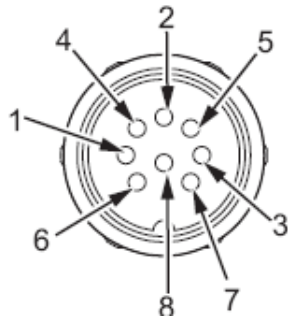
** To be used with 3 meter serial cable to provide a 9 meter serial cable.*

RS-232 Connector for Videojet 1610 DH

When the communication port is configured to RS-232, the software handshaking and hardware handshaking is available for both communication port 1 and port 2. Three cables are available for communication channels - I/O cable (RS 232 - 3 m), I/O cable (RS-232 -6 m) and I/O cable (RS 232 / RS 485 - 6 m).



RS 232 Connector (3 m)



RS 232 Connector (6 m)

The table below shows the pin connection information for RS-232 software handshaking.

8 Pin DIN Plug		DB 9 Pin Connector	
DIN Pins	Function	DB 9 Pins	Function
1	TXD (Transmit Data)	2	Received Data
3	RXD (Received Data)	3	Transmit Data
8	Gnd	5	System Ground

RS-232 Pin Connection Information - Software Handshaking

The table below shows the pin connection information for RS-232 hardware handshaking.

8 Pin DIN Plug		DB 9 Pin Connector	
DIN Pins	Function	DB 9 Pins	Function
1	TXD (Transmit Data)	2	Received Data
3	RXD (Received Data)	3	Transmit Data
8	Gnd	5	System Ground
2	RTS (Request to Send)	8	Clear to Send
4	CTS (Clear To Send)	7	Request to Send
		1+4+6	
		1	Data Carrier Detect
		4	Data Terminal Ready
		6	Data Set Ready

RS-232 Pin Connection Information - Hardware Handshaking

Part Number

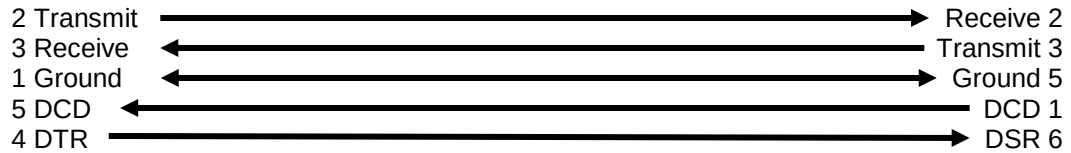
The mating cable (male, 8 pin DIN connector - P/N: 500-0036-584).

The mating cable (male, 8 pin DIN connector - P/N: 500-0036-584).

RS-232 Cable Pin Out (All 1000 series except Videojet 1610DH)

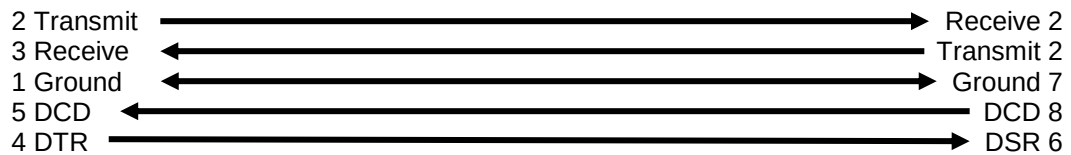
DIN TO DB9

DIN CONNECTOR (5 PIN) PRINTER SIDE **DB9**



DIN TO DB25

DIN CONNECTOR (5 PIN) PRINTER SIDE **DB25**



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5. Appendix

See table below showing CMD's for different variants.

1510 Name	1510CMD Number	1610 Name	1610CMD Number	1710 Name	1710CMD Number	1610 DH Name	1610DHCMD Number
Set Module Widths	B	Set Module Widths	B	Set Module Widths	B	Set Module Widths	B
Delete Message	C	Delete Message	C	Delete Message	C	Delete Message	C
Clear User Field Data	D	Clear User Field Data	D	Clear User Field Data	D	Clear User Field Data	D
Request Error Status	E	Request Error Status	E	Request Error Status	E	Request Error Status	E
Get Print/Product Counters	G	Get Print/Product Counters	G	Currently not supported	G	Currently not supported	G
Request Part Number	H	Request Part Number	H	Request Part Number	H	Request Part Number	H
Start Jet (Disabled)	J	Start Jet (Disabled)	J	Start Jet (Disabled)	J	Start Jet (Disabled)	J
Stop Jet	K	Stop Jet	K	Stop Jet	K	Stop Jet	K
Update Logo Data	L	Update Logo Data	L	Update Logo Data	L	Update Logo Data	L
Message Select	M	Message Select	M	Message Select	M	Message Select	M
Print On/Off	O	Print On/Off	O	Currently not supported	O	Currently not supported	O
Modify Message parameters	P	Modify Message parameters	P	Modify Message parameters	P	Modify Message parameters	P
Request Current Message Name	Q	Request Current Message Name	Q	Currently not supported	Q	Currently not supported	Q
Reset Print/Product	R	Reset Print/Product	R	Currently not supported	R	Currently not supported	R

Counters		Counters					
Update Message Text	T	Update Message Text	T	Update Message Text	T	Update Message Text	T
Update User Field Data	U	Update User Field Data	U	Update User Field Data	U	Update User Field Data	U
Set Date/Time	Z	Set Date/Time	Z	Currently not supported	Z	Currently not supported	Z
Command Successful Response	\$	Command Successful Response	\$	Command Response	\$	Command Response	\$
Command Error Response	!	Command Error Response	!	Currently not supported	!	Currently not supported	!